Did the CHIPS Act and the Inflation Reduction Act cause the US factory construction boom?

A difference-in-differences and synthetic control analysis

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1 Introduction

In August 2022 the United States passed two industrial policy laws within a week. The CHIPS and Science Act (9 August) funds the return of semiconductor production with about $52bn of subsidies and tax credits. The Inflation Reduction Act, or IRA (16 August), grants large tax credits to clean energy and to its manufacturing supply chain: batteries, solar panels, electric vehicles and hydrogen. In the following two years, real spending on US factory construction rose from $63.8bn in 2021 to $175.2bn in 2024 (annual averages, 2019 dollars). Over the same period, however, firms were reshoring production after COVID, a global chip shortage was pushing chipmakers to expand, demand for AI chips exploded and the Federal Reserve raised its rates. A simple before and after comparison cannot separate these forces.

Our question is the following: **how much of the increase in factory construction spending in the computer, electronic and electrical sector since August 2022 was caused by the CHIPS Act and the IRA?** The treatment is the package formed by the two laws, which entered into force the same month and cannot be separated in the data. The outcome is real private spending on the construction of plants in that sector, the sector of semiconductor, electronics and battery plants. The population is US manufacturing, split into 7 subsectors, observed monthly from July 2019 to August 2026. A second analysis widens the population to 21 OECD countries to ask whether the effect is visible in the aggregate investment of the United States.

The results fit in three sentences. Compared with the other manufacturing subsectors, the laws raised factory construction in the treated sector by +189% to +452%, i.e. $43.0bn to $53.9bn per year in 2019 dollars; the range comes from a rise in late 2021, before the laws, that the data cannot attribute with certainty. The effect peaks in late 2023 and then declines as plants are completed. At the country level, US non-residential investment exceeds its synthetic twin by +8.3 index points, but this gap does not stand out from placebo countries and almost vanishes without Ireland among the donors.

2 Data

Table 1 lists the sources. The main outcome comes from the Census Bureau series on the value of private construction put in place (U.S. Census Bureau 2026). It measures spending actually incurred each month, not announced projects, at a seasonally adjusted annual rate. Since construction costs rose sharply in 2021 and 2022, nominal amounts are deflated by the construction materials price index from FRED (Federal Reserve Bank of St. Louis 2026), normalised to its 2019 average, and are expressed in 2019 dollars. The difference-in-differences works on the logarithm of this real spending, so that effects read as percentages.

For the country comparison, we use OECD quarterly national accounts on gross fixed capital formation in non-residential buildings and structures (OECD 2026). These series are already in volume, but in national currencies, so each country is rebased to 100 = 2019. The Clean Investment Monitor (Rhodium Group and MIT Center for Energy and Environmental Policy Research 2026) is used only as context: announced investment in clean technology factories peaked in 2022 at $90.8bn (2024 dollars), the year of the IRA, and actual spending followed one to two years later.

Table 1: Data sources. All series are public and were downloaded on 26 October 2026.

Source	Series	Frequency	Coverage	Role
Census Bureau	Private construction spending: 7 manufacturing subsectors, offices, commercial	Monthly	2010-01 to 2026-08	DiD outcome
FRED	Construction materials price index (WPUSI012011)	Monthly	2010-01 to 2026-09	Deflator
OECD	Investment in non-residential structures, volume, 21 countries	Quarterly	2010-Q1 to 2026-Q2	Synthetic control outcome
Clean Investment Monitor	Clean technology factories, announced and actual investment	Quarterly	2018-Q1 to 2026-Q2	Context

3 Identification strategy

3.1 Causal graph and estimand

The difference-in-differences (DiD) compares the change in spending in the treated sector with the change in the other manufacturing subsectors, called the control sectors, before and after August 2022. Figure 1 shows what this double comparison removes. Stable differences between sectors (α_s, such as size or capital intensity) disappear when each sector is compared with itself over time. Shocks common to all sectors (λ_t, such as interest rates or materials prices) disappear when the treated sector is compared with the control sectors. What remains is a factor that evolves differently across sectors (U_{st}), such as the 2021 chip shortage or AI-driven demand: it affects both the timing of the treatment, since the laws target this sector precisely because chips became strategic, and the outcome. This is the main threat to the design.

The estimand is the average treatment effect on the treated (ATT): the average difference, from August 2022 onwards, between the log spending observed in the treated sector and the log spending it would have recorded without the laws. We call this difference δ and report it as a percentage, $e^{\delta} - 1$. The synthetic control estimates the same kind of quantity for the United States as a whole, in index points.

3.2 Assumptions and their credibility

Three assumptions are needed. Consistency requires a well-defined treatment: here it is the package of the two laws, and we do not try to separate them. No anticipation requires that nothing reacts before August 2022. This

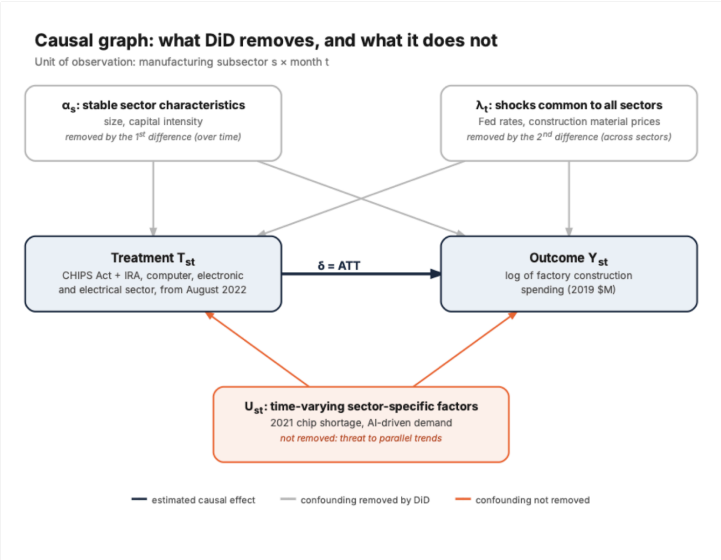


Figure 1: Causal graph of the difference-in-differences. Grey arrows are confounders removed by the design, orange arrows are confounders it does not remove.

is doubtful, since the CHIPS Act was debated in Congress from 2021 and large chipmakers announced US plants before the vote. Parallel trends, the key assumption, states that without the laws the treated sector would have evolved like the control sectors, in percentage terms. It cannot be tested after 2022, but it can be checked before: if the two groups moved together for several years, the assumption is more plausible. Section 4.2 shows that this holds from mid-2019 to mid-2021, but not over a longer window. Finally, the control sectors must not be affected by the laws themselves. This is mostly true, with one exception: transportation equipment includes electric vehicle plants, which benefit from the IRA.

3.3 Choice of control group and of a second design

Our 6 control sectors (chemicals, food and beverages, plastics, non-metallic minerals, fabricated metals and transportation equipment) come from the same survey, face the same interest rates, labour market and materials prices, and are not the direct target of the laws. Offices and commercial buildings serve as alternative controls in the sensitivity analysis. With a single treatment date, the regression with sector and month fixed effects is exactly the simple comparison of means, so the problems of DiD with staggered treatment timing (Goodman-Bacon 2021) do not arise.

The DiD fails if the treated sector was hit by its own shock at the same time as the laws. The chip shortage and the AI boom are such shocks, but they were global. This motivates a second design, the synthetic control (Abadie et al. 2010; Abadie 2021), which compares the United States with a weighted average of other OECD countries exposed to the same global shocks but not to the US laws. Its own weaknesses are different: its outcome covers all non-residential structures, so the effect is diluted, and the comparison countries may have been affected by the laws through relocated projects, or by their own shocks such as the European energy crisis. The two designs therefore fail for different reasons, and agreement between them would be more convincing than either alone.

4 Difference-in-differences

4.1 The raw comparison

Figure 2 plots monthly real spending in the treated sector and the average of the control sectors. The rise starts before the laws. Treated-sector spending goes from \$8,020M in June 2021 to \$22,649M in June 2022, with a jump between September and October 2021 (from \$8,369M to \$12,967M), while the control average barely moves (from \$6,653M to \$7,016M). This rise can reflect anticipation of the laws or the chip shortage. The data cannot tell which.

4.2 Event study and choice of the before window

An event study computes, quarter by quarter, the gap in log spending between the treated sector and the control sectors, relative to its average over a reference period. If parallel trends held before the laws, the points before 2022 should stay close to zero. Figure 3 shows three phases. From 2015 to 2019, the treated sector catches up with

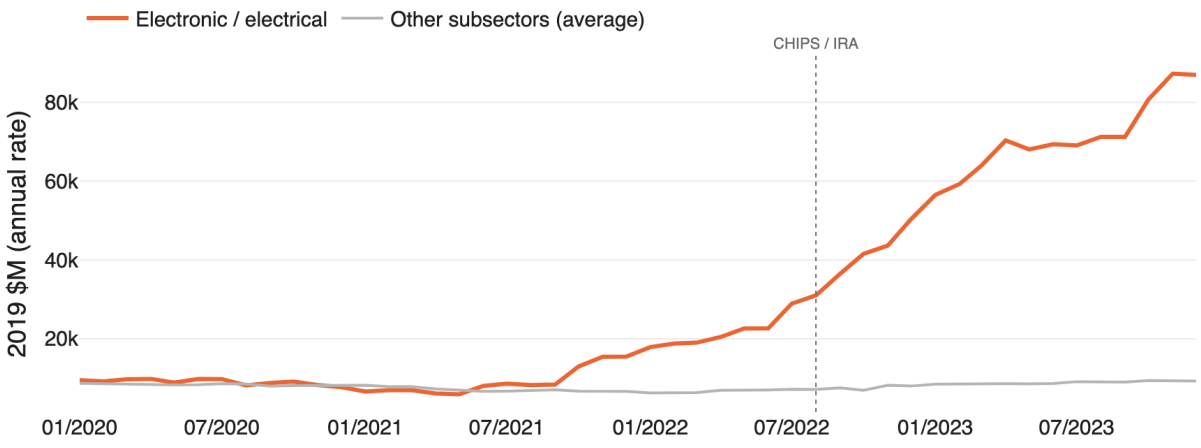


Figure 2: Monthly construction spending, treated sector and average of the control sectors, 2019 \$M at an annual rate. The dotted line marks August 2022.

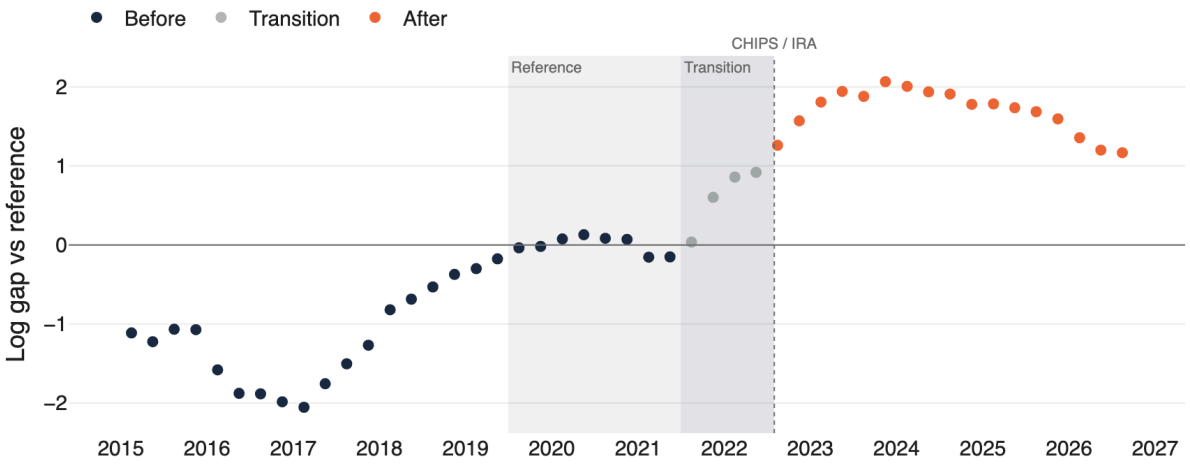


Figure 3: Event study: log gap between the treated sector and the control sectors by quarter, relative to its mean from July 2019 to June 2021 (light band). The darker band is the transition period excluded from the estimation.

the control sectors (−2.05 in 2017-Q1), so a long before window would violate parallel trends. From mid-2019 to mid-2021, the gap stays between −0.15 and +0.13. From 2021-Q4 it rises (+0.60), before the laws.

We therefore use this plateau as the before period (24 months, July 2019 to June 2021) and exclude the 13 transition months up to July 2022. This window was chosen after looking at the data, which carries a risk of choosing a favourable reference; Section 6 reports the result with the full window from 2015.

4.3 Estimates

Between the before period and the after period (49 months, August 2022 to August 2026), log spending rises by 2.007 in the treated sector and by 0.298 in the control sectors, hence $\delta = 1.709$. The regression with sector and month fixed effects gives the same value.

The rise of late 2021 leads to two readings, shown in Table 2. The upper bound assumes that, without the laws, the sector would have stayed at its 2019 to 2021 level, so that all of the late-2021 rise is anticipation of the laws. The lower bound assumes that this rise, linked to the chip shortage, would have happened anyway, and uses the transition months as the reference. The truth probably lies in between. In dollars, the effect is the observed spending minus the spending predicted without the laws, Y/e^{δ}; the two bounds are closer in dollars than in percent because they start from the same observed spending.

The effect is not constant. Measured against the plateau, it reaches +253% in 2022-Q3, peaks at +689% in 2023-Q4, then falls back to +221% in 2026-Q3. Since the outcome is a flow of construction spending, this decline is expected: once plants are built, spending stops. It does not mean that the plants disappear. These dynamics use the plateau as reference and are therefore upper-bound figures.

4.4 Inference

With a single treated sector, the usual standard error clustered by sector (0.094) is not reliable, so uncertainty is assessed with placebos: the same estimate computed where no effect should exist. In-time placebos apply the same DiD, with the same window lengths, to 53 fake treatment dates between February 2013 and June 2017, before the 2021 rise. The largest of them is 0.66, and none of them reach the true δ (Figure 8 in the appendix). The true δ exceeds it by 1.05 with the upper bound and by 0.40 with the lower bound. Because consecutive fake dates share most of their months, they are not independent draws, and the count is not a p-value; it shows that

Table 2: Range of the DiD effect, average since August 2022. Effect in % is $e^{\delta} - 1$; dollars are 2019 $bn per year.

Reading	Reference window	δ (log)	Effect	$bn per year
Upper bound (anticipation)	2019-07 to 2021-06	1.709	+452%	$53.9bn
Lower bound (chip shortage)	2021-07 to 2022-07	1.061	+189%	$43.0bn

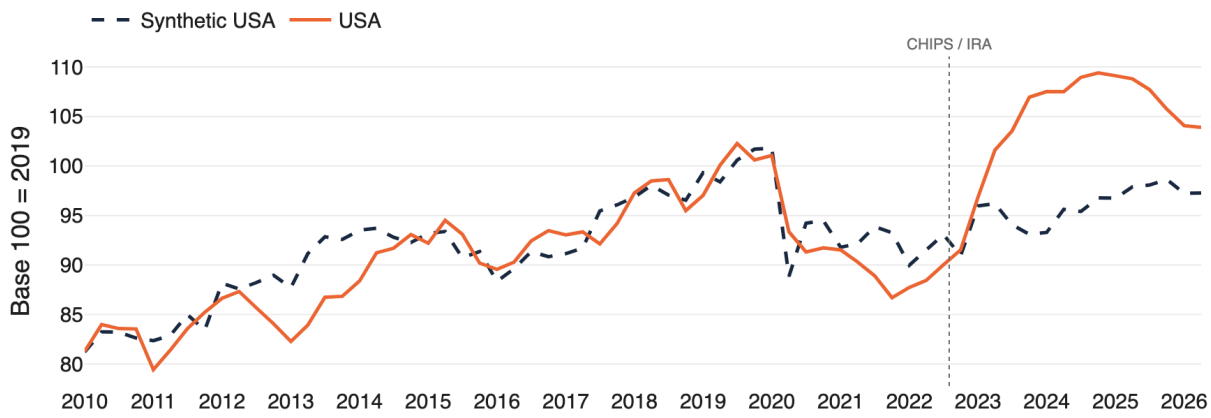


Figure 4: Investment in non-residential buildings and structures, United States and synthetic twin, volume index 100 = 2019. The weights are fitted on 2015-Q1 to 2022-Q2.

a gap of this size never occurred in the sector's recent history. The permutation test assigns the treatment to each of the 7 sectors in turn. The treated sector has the only positive δ (placebos from −0.677 to −0.089), which gives p = 0.143, i.e. 1/7, the smallest value possible with 7 sectors. A conventional 5% threshold is out of reach by construction.

5 Synthetic control

5.1 Donor weights and fit

The synthetic twin of the United States is a weighted average of the 20 other OECD countries, called donors, with non-negative weights summing to one, chosen to track the US index as closely as possible from 2015-Q1 to 2022-Q2. The weights select 8 donors: Ireland (0.27), Japan (0.15), Sweden (0.13), Canada (0.13), Finland (0.13), Australia (0.09), Spain (0.07), Czechia (0.04) (Figure 7 in the appendix). European countries account for 64% of the weight.

Figure 4 compares the United States with its twin. The fit is good until 2019, then weaker: the root mean squared prediction error (RMSPE) before treatment is 2.44 index points, and the United States falls 6.6 points below its twin in 2021-Q4. After 2022-Q3, the United States exceeds its twin by +8.3 points on average, with a peak of +14.2 in 2024-Q1 and +6.6 in 2026-Q2. Part of this gap may simply be a rebound from the 2021 trough.

5.2 Inference with placebos in space

Placebos in space treat each donor country in turn as if it had passed the laws, build its own twin from the other donors, and compare the post-treatment divergence with its pre-treatment fit (the ratio of post to pre RMSPE). Figure 5 shows that several countries diverge from their twin as much as the United States after 2022. The United States ranks 6th out of 21, behind Italy, France, Austria, Spain, the Netherlands, which gives a rank p-value of 0.286 (the minimum would be 1/21). Keeping only the 14 countries whose pre-treatment fit is at most twice as poor as that of the United States raises it to 0.429. The US gap does not stand out.

5.3 Dependence on Ireland

Ireland carries the largest weight (0.27) although its series is very volatile: its correlation with the United States before 2022 is the highest of all donors (0.67), but its mean squared deviation is large (10.06 points). Removing Ireland from the donors reduces the mean gap from +8.3 to +1.7 points, while removing any other donor leaves it between +7.7 and +9.2. The country-level result rests on a single donor.

6 Sensitivity analysis and limitations

Figure 6 summarises the sensitivity of both designs. In the DiD, excluding transportation equipment from the controls or adding offices and commercial buildings keeps the effect between +430% and +495%. Measuring the outcome in levels gives +591%, but assumes that the treated sector would have gained as many dollars as the control sectors, rather than the same percentage. The before window starting in 2015 gives +915%, because it

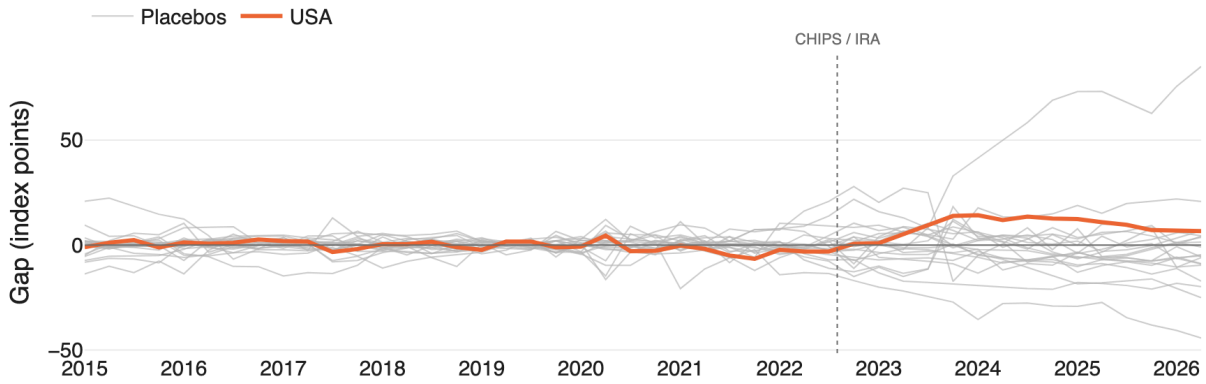


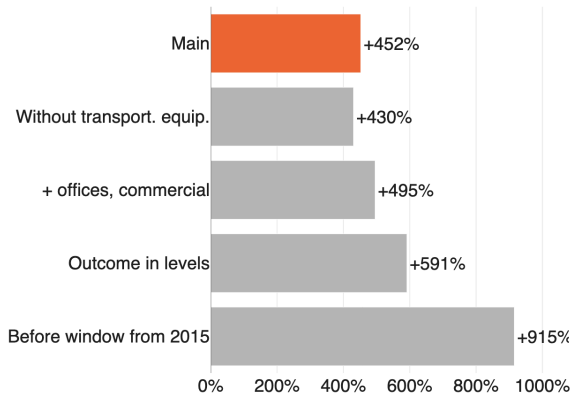
Figure 5: Gap between each country and its synthetic twin, in index points. Orange: United States. Grey: the other countries treated as placebos.

mixes the 2017 to 2019 catch-up with the effect: the result depends strongly on the reference period. In the synthetic control, a fake treatment in 2019-Q1 gives −2.2 points and moving the treatment to 2021-Q3 gives +4.1; Ireland remains the decisive donor.

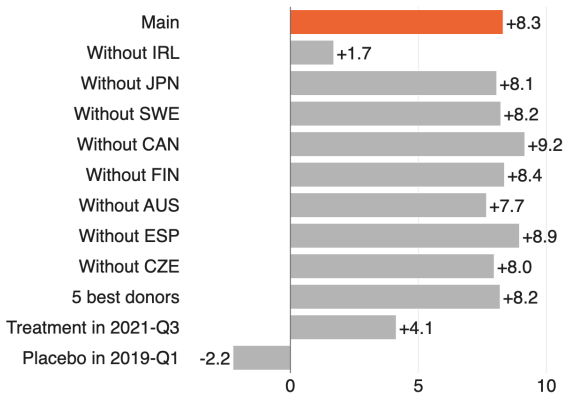
Table 3 lists the remaining threats and the direction in which each would bias the estimates.

Table 3: Main threats to identification and direction of the bias.

Threat	Why it matters	Likely bias
Anticipation	Plants announced while the CHIPS Act was debated in 2021 raise spending before August 2022.	Lower bound too low; covered by the range.
Chip shortage (selection on trends)	The laws target the sector that was already expanding because of a sector-specific shock.	Upper bound too high.
AI boom	Demand for AI chips raises fab construction from 2023, independently of the laws.	DiD overestimated.
Spillover to transportation equipment	Electric vehicle plants in this control sector also receive IRA credits.	DiD underestimated (small: removing it changes little).
Spillover to Europe	Projects attracted to the United States would otherwise have been built in donor countries.	Twin too low, synthetic control overestimated.
European energy crisis	Most of the twin's weight is European, and Europe slowed down in 2022-2023.	Synthetic control overestimated.
Outcome too broad for the synthetic control	Offices and infrastructure weigh much more than the plants targeted by the laws.	Synthetic control effect diluted, underestimated.
A single treated sector	Inference rests on a handful of sectors and on placebos, not on sampling uncertainty.	No direction; limits the precision of any claim.



(a) DiD: average effect since August 2022, $e^{\delta} - 1$.



(b) Synthetic control: mean gap after 2022-Q3, index points.

Figure 6: Sensitivity of the two designs. Orange bars are the main specifications.

7 Conclusion

What can be claimed is that factory construction in the computer, electronic and electrical sector rose far more after August 2022 than in any other manufacturing subsector, by +189% to +452%, i.e. $43.0bn to $53.9bn per year in 2019 dollars. No comparable gap appears at any earlier date or in any other sector. The size of the effect, rather than a formal test, is the main evidence.

What cannot be claimed is that all of this increase is due to the laws. The DiD cannot separate the laws from the chip shortage, which started before them, or from the AI boom, which followed them. At the country level the effect is not distinguishable from the variation seen in other countries, and it depends on a single donor. A broad national aggregate is however a weak test of a policy aimed at one type of plant.

Going further would require finer data. A state-level or plant-level panel, such as the Clean Investment Monitor's project database, would allow comparing places more or less exposed to the credits, and sectors targeted by the IRA alone (batteries, solar) would offer additional treated units. The partial rollback of some credits from 2025 also provides a second event to test whether investment responds to the policy in the opposite direction.

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Appendix

Reproducibility

All figures and numbers in this report are computed when the document is rendered, from the functions in `src/` and the files in `data/processed/`.

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Additional figures

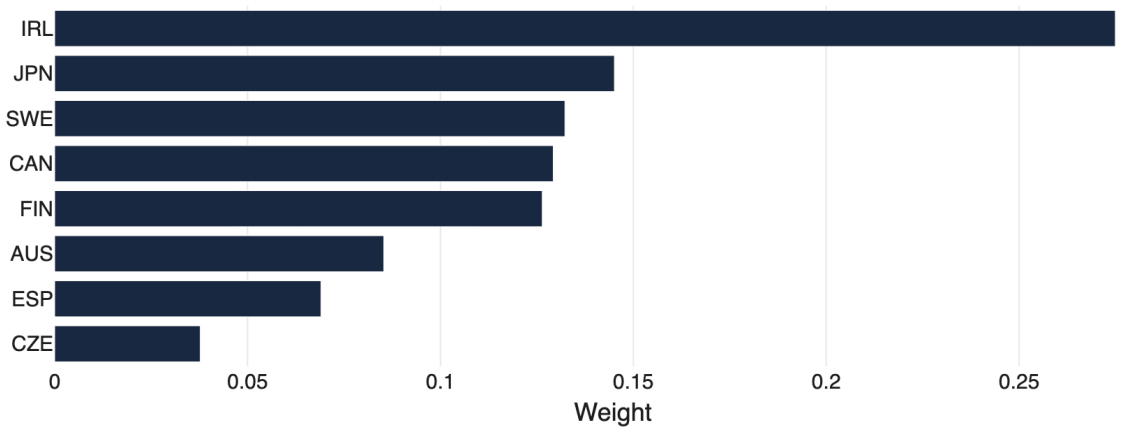


Figure 7: Donor weights in the synthetic twin, fitted on 2015-Q1 to 2022-Q2 (non-zero weights only).

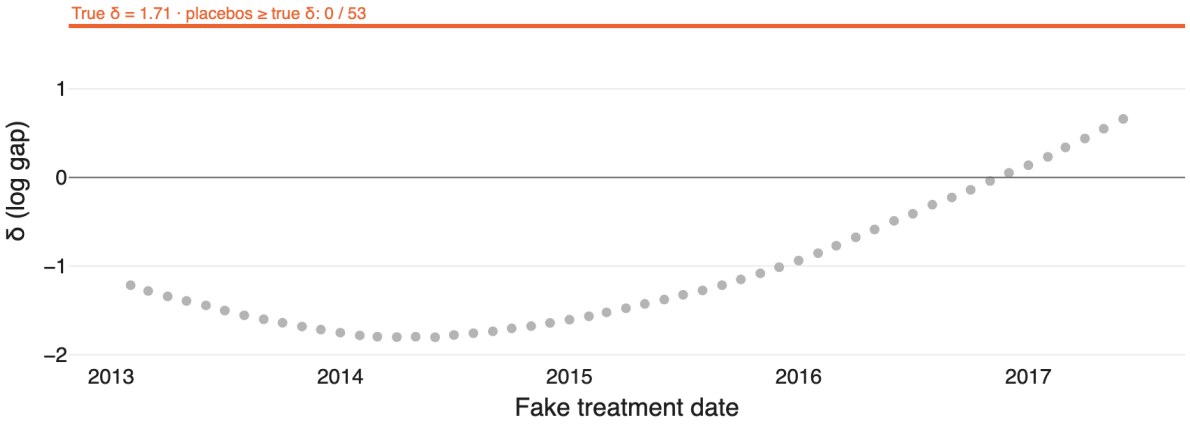


Figure 8: In-time placebos: the same DiD, with the same window lengths, estimated at fake treatment dates, against the true δ (orange line).